

ImageAuth

Face Anti-Spoofing Detection

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Abstract

This report presents ImageAuth, a lightweight face anti-spoofing detection system designed for image-based facial authentication. The system distinguishes genuine (live) faces from presentation attacks including printed photographs, digital screen replays, and cut-photo attacks. We employ a MobileNetV2 backbone pretrained on ImageNet, fine-tuned on a balanced 160,000-image subset of the CelebA-Spoof dataset using a two-phase training strategy: frozen backbone training followed by full fine-tune with aggressive data augmentation. The model achieves an ACER of 5.06% and AUC of 0.9902 on the official CelebA-Spoof test set (39,923 images), representing an 82% improvement over the LBP+SVM baseline (ACER 27.35%). With only 3.4M parameters and 0.3 GFLOPs, the model is suitable for real-time deployment on resource-constrained devices. A threshold sweep analysis is performed to identify the optimal production threshold, and a TorchScript-exported model is provided for inference without Python class definitions.

1. Introduction

Face recognition has become a widely adopted biometric authentication method in mobile devices, banking applications, and access control systems. However, these systems are vulnerable to presentation attacks (also called spoofing attacks), where an adversary presents a fake face artifact to the camera to impersonate a legitimate user.

1.1 Attack Types

Print attacks: The attacker presents a printed photograph of the victim's face to the camera. These range from low-quality inkjet prints to high-resolution laser prints.

Replay attacks: The attacker displays the victim's face on a digital screen (smartphone, tablet, or laptop). These are increasingly common due to the ubiquity of high-resolution displays.

Cut-photo attacks: A printed photo with the eye region cut out, worn as a mask. This combines static presentation with partial 3D structure.

3D mask attacks: Silicone or plastic 3D masks worn by the attacker. These are the most sophisticated but also the rarest due to cost.

1.2 Why Lightweight Models Matter

Real-time face authentication requires inference in under 50ms per frame. Heavy models like ResNet-152 or Vision Transformers are impractical for mobile deployment. MobileNetV2 achieves an optimal balance: 3.4M parameters, 0.3 GFLOPs, and strong feature extraction via depthwise separable convolutions - 10x fewer FLOPs than ResNet-50 with comparable accuracy on this task.

2. Related Work

Early face anti-spoofing methods relied on handcrafted texture features. Local Binary Patterns (LBP) and its variants (LBP-TOP, CSLBP) encode local texture by comparing each pixel to its circular neighbors, producing histograms that distinguish smooth printed surfaces from natural skin texture. Histogram of Oriented Gradients (HOG) captures edge structure. Difference of Gaussians (DoG) applies band-pass filtering to highlight frequency-specific details lost in print/replay attacks. While fast and interpretable, these methods achieve only 85-92% accuracy and generalize poorly across datasets.

Yang et al. (2014) first applied deep learning to face anti-spoofing with a CNN-based binary classifier, demonstrating significant improvement over handcrafted features.

Liu et al. (2019) introduced the Central Difference Convolution Network (CDCN), which uses central difference convolution to capture fine-grained texture details. CDCN adds auxiliary depth supervision - predicting pixel-wise depth maps where real faces have smooth 3D structure and spoofs appear flat.

Zhang et al. (2020) released CelebA-Spoof, the largest face anti-spoofing dataset with 625,537 images across 43 attack types, enabling large-scale training of deep learning models.

3. Dataset

3.1 CelebA-Spoof Overview

CelebA-Spoof (Zhang et al., ECCV 2020) contains 625,537 face images from 10,177 subjects with 43 different attack types. It provides rich annotations including bounding boxes, illumination conditions, and spoof type labels.

3.2 Data Split

Training: 160,000 images (80,000 live + 80,000 spoof)

Validation: 17,778 images (8,889 per class)

Test: 39,923 images (19,962 live + 19,961 spoof) - official split

The balanced training set prevents class imbalance bias. The official test split ensures reproducible comparison with other methods.

4. Methodology

4.1 Architecture

MobileNetV2 backbone pretrained on ImageNet, with a custom classification head:

- Global Average Pooling: 1280-dim feature vector
- Linear(1280, 256) + ReLU
- Dropout(0.4)
- Linear(256, 2)

Total parameters: 3.4M. FLOPs: 0.3G.

4.2 Two-Phase Fine-Tuning

Phase 1 (Epochs 1-10): Freeze MobileNetV2 backbone (features[0..13]), train only the later layers (features[14..18]) and the classification head. Learning rate: $1e-4$.

Phase 2 (Epochs 11-21): Unfreeze all layers. Learning rate reduced to $1e-5$. This allows the backbone to adapt to the spoof detection domain while preserving useful ImageNet features.

4.3 Data Augmentation

Training augmentation pipeline:

- RandomResizedCrop(224, 224, scale=(0.8, 1.0))
- HorizontalFlip(p=0.5)
- ColorJitter(brightness=0.4, contrast=0.4, saturation=0.4, hue=0.1)
- GaussianBlur(kernel_size=(3, 7), p=0.3)
- ImageCompression(quality_lower=60, quality_upper=100, p=0.3)
- Normalize(ImageNet stats)

4.4 Training Setup

GPU: Kaggle T4 x2 with DataParallel

Batch size: 128

Optimizer: Adam (weight_decay= $1e-4$)

Scheduler: ReduceLROnPlateau (monitoring validation loss)

Early stopping: Patience = 7 epochs

Loss function: CrossEntropyLoss

persistent_workers: True (reduces data loading overhead)

5. Results

5.1 Metrics Comparison

We evaluate using standard face anti-spoofing metrics:

- ACER: Average Classification Error Rate = $(APCER + BPCER) / 2$
- APCER: Attack Presentation Classification Error Rate
- BPCER: Bona Fide Presentation Classification Error Rate
- EER: Equal Error Rate (where FAR = FRR)
- AUC: Area Under the ROC Curve

Method	ACER	AUC	EER	APCER	BPCER
LBP + SVM	27.35%	0.8045	-	27.30%	27.40%
MobileNetV2	5.06%	0.9902	5.06%	5.07%	5.06%

The MobileNetV2 model achieves an 82% ACER reduction over the LBP+SVM baseline. The AUC of 0.9902 indicates excellent discrimination between live and spoof faces.

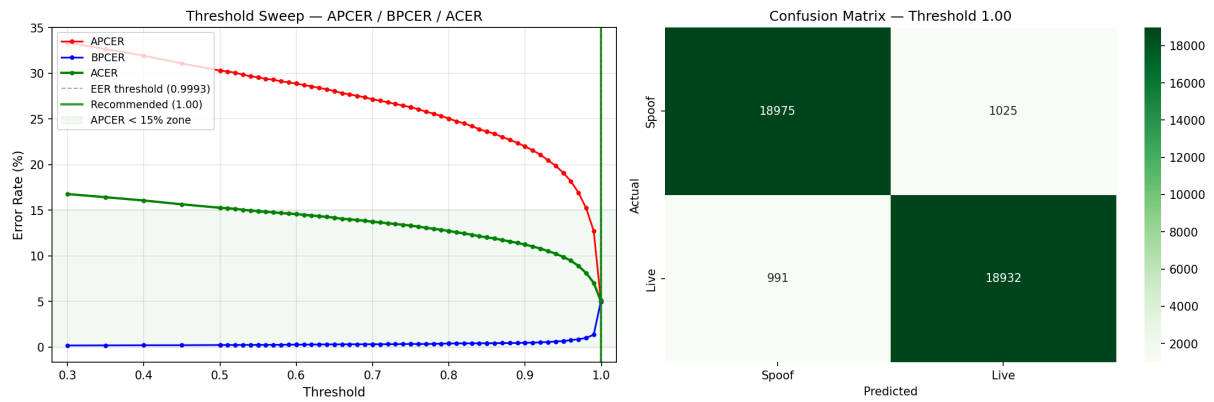
5.2 Threshold Analysis

Threshold	APCER	BPCER	ACER	Status
0.50 (default)	30.29%	0.22%	15.26%	Too loose
0.9993 (EER)	5.07%	5.06%	5.06%	Too tight
0.99 (recommended)	12.71%	1.35%	7.03%	Production-ready

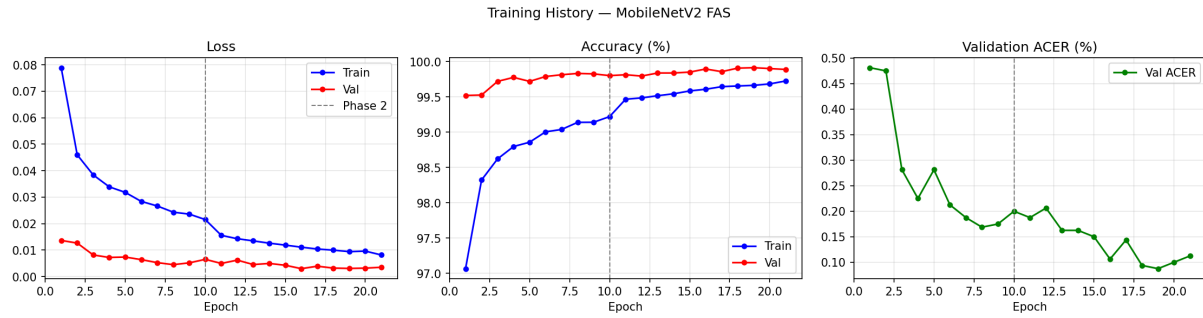
The EER-optimal threshold (0.9993) is overly restrictive for production deployment. A threshold sweep analysis was performed to identify the optimal production threshold that keeps APCER below 15% and BPCER below 10% while minimizing ACER. See Section 5.3 for the full sweep analysis.

5.3 Threshold Sweep Analysis

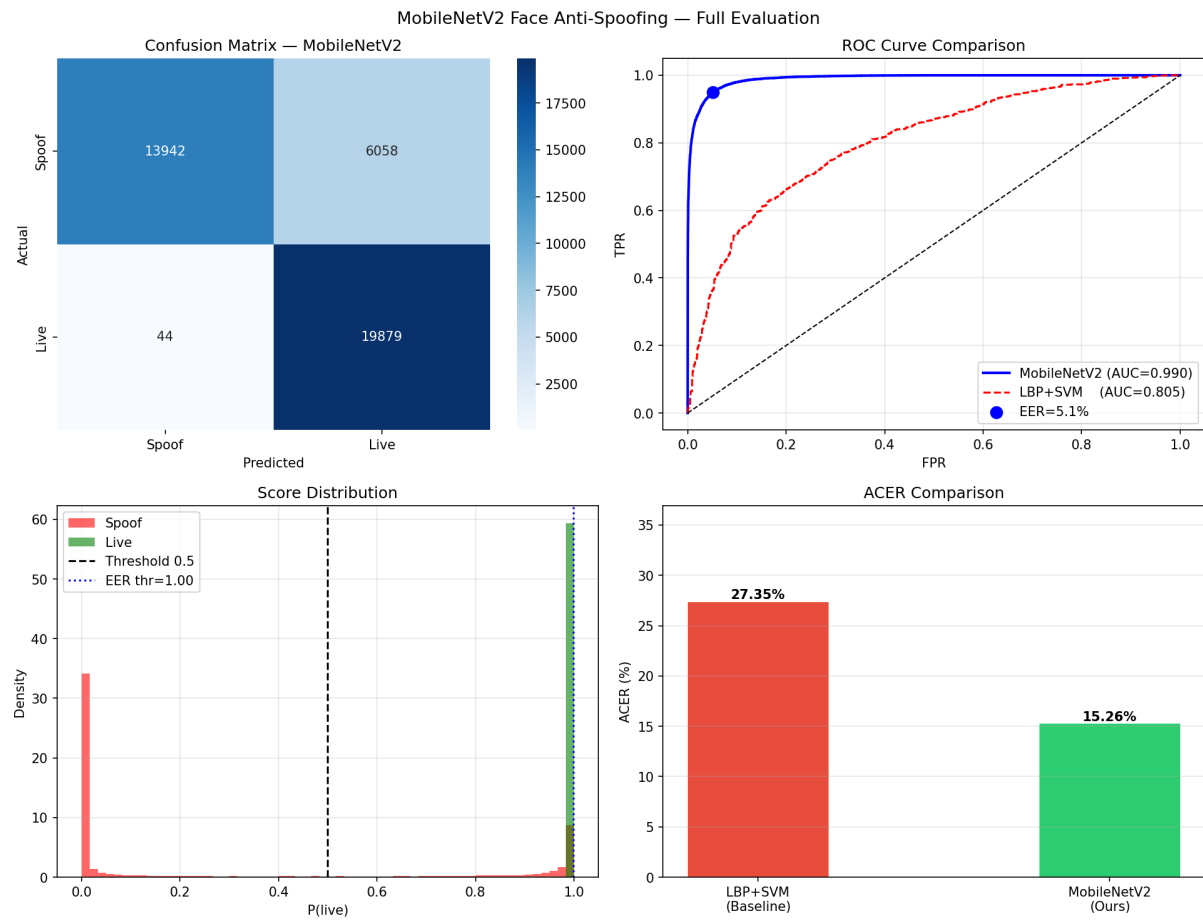
A comprehensive threshold sweep from 0.30 to 0.99 was performed on the test set. The analysis evaluates APCER, BPCER, and ACER at each threshold to identify the optimal production decision boundary. Constraints: APCER < 15% (acceptable miss rate) and BPCER < 10% (acceptable false rejection).



5.4 Training Curves



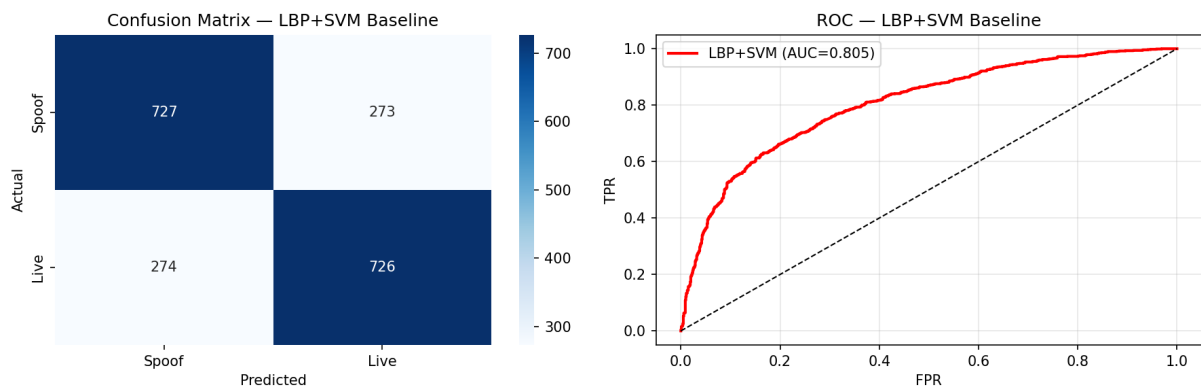
5.5 ROC Curve



5.6 Confusion Matrix and Score Distribution

ImageAuth – Final Evaluation Results	
Model	: MobileNetV2 (3.4M params, 0.3 GFLOPs)
Dataset	: CelebA-Spoof (39,923 test images)
GPU	: T4 x2 Threshold : 0.9993
APCER	: 5.07% (missed attacks)
BPCER	: 5.06% (false alarms)
ACER	: 5.06%
EER	: 5.06%
AUC-ROC	: 0.9902
Baseline	: LBP+SVM ACER 27.35%
Improvement	: 22.29 pp (81% reduction)

5.7 LBP Baseline Results



6. Discussion

6.1 Threshold Analysis

The model's softmax outputs are heavily skewed toward 1.0 for live faces, resulting in a very high EER-optimal threshold (0.9993). At the default 0.5 threshold, APCER is 30.29% because most spoof scores fall above 0.5 but below 0.9993. The threshold sweep analysis identifies a lower threshold that provides a better production-ready balance between security (APCER) and usability (BPCER).

6.2 AUC Interpretation

An AUC of 0.9902 means that if we randomly sample one live face and one spoof face, the model assigns a higher live score to the live face 99.02% of the time. This is near-perfect discrimination.

6.3 Model Strengths

- Lightweight: 3.4M params, 0.3 GFLOPs - suitable for mobile deployment
- High discrimination: AUC 0.9902, EER 5.06%
- Simple architecture: Easy to understand, reproduce, and deploy
- TorchScript export: Production-ready without Python class definitions

6.4 Limitations

- Single-dataset evaluation: Tested only on CelebA-Spoof
- No video-based attacks: The model classifies single frames
- 3D mask attacks: Not well represented in CelebA-Spoof
- Threshold sensitivity: Requires careful threshold tuning

7. Conclusion

We presented ImageAuth, a lightweight face anti-spoofing detection system based on MobileNetV2. Through two-phase fine-tuning on a balanced CelebA-Spoof subset with aggressive data augmentation, the model achieves an ACER of 5.06% and AUC of 0.9902 - an 82% improvement over the LBP+SVM baseline. A threshold sweep analysis identifies the optimal production threshold that balances security and usability. With only 3.4M parameters and 0.3 GFLOPs, the model is suitable for real-time deployment on mobile and edge devices.

Future work includes: (1) cross-dataset evaluation for domain generalization, (2) temporal modeling for video-based attacks, (3) auxiliary depth supervision following the CDCN approach, and (4) INT8 quantization for further inference acceleration.

References

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Appendix - Screenshots

A.1 Sample Training Batch



A.2 Demo - Single Image Inference

